

YYB Quant Platform: A Systematic Framework for Alpha Factor Mining, Classification, and Multi-Strategy Combination

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Abstract

We present the YYB Quantitative Research Platform, an integrated system for alpha factor mining, economic classification, and multi-strategy portfolio construction in the Chinese A-share market. The platform ingests 89 daily financial fields spanning 5,515 stocks over 970 trading days (2020–2023) and supports factor expression parsing, single-factor backtesting, and four distinct combination strategies (Equal-Weight, ICIR-Weighted, Ridge-Regularized, and LightGBM). A key innovation is the automatic economic classification of 761 single factors into five interpretable groups—Momentum, Reversal, Microstructure, Liquidity, and Volatility—with an additional 512 factors tagged as “mixed” due to their multi-dimensional nature. Through systematic experiments across 55+ combinations, we demonstrate that (1) mixed-style factors, previously excluded as unclassifiable, achieve an average combination Sharpe of 9.11, significantly outperforming clear-group factors (7.43); (2) LightGBM with a hybrid mix-clean factor set achieves a Sharpe ratio of 12.51 with IC=0.051; (3) ICIR-weighted combination consistently outperforms equal-weight for small factor sets ($N \leq 5$); and (4) style diversification from cross-group selection yields superior risk-adjusted returns. We submit 10 final factors with pairwise correlations below 0.5 for competition evaluation.

1 Introduction

Alpha factor mining in the Chinese A-share market presents unique challenges: high stock count (5,515), limited trading history (970 days), and complex market microstructure. Traditional approaches either rely on manual factor construction with limited coverage, or systematic enumeration that generates highly correlated signals. We address this gap through the YYB Platform, which combines (1) a flexible expression parser supporting 89 data fields and 20+ operators, (2) automatic economic classification of factors into interpretable groups, (3) four complementary combination strategies with rigorous out-of-sample evaluation, and (4) a web-based research dashboard enabling interactive exploration.

The platform’s design philosophy centers on three principles: *economic interpretability*—every factor must be traceable to a specific market mechanism; *memory safety*—heavy computations run in isolated subprocesses with guaranteed memory release; and *reproducibility*—all backtests use identical data windows and evaluation metrics.

Our experiments reveal a counterintuitive finding: factors that the classifier labels as “mixed” (unable to assign to a single economic group) constitute the **most powerful input set for portfolio combination**. These mixed factors, representing 67% of all factors, achieve an average combination Sharpe ratio of 9.11—22.6% higher than the best single-style factors and 47% higher than style-balanced portfolios. We attribute this to their inherent multi-dimensionality: by design, mixed factors capture signals spanning multiple economic mechanisms, providing natural diversification within each factor.

2 Platform Architecture

2.1 Data Pipeline

The platform ingests daily frequency data for 5,515 A-share stocks across 970 trading days (2020-01-02 to 2023-12-29). The DataPipeline module provides:

- 89 base data fields including OHLCV, returns, volume, volatility, and derived metrics
- Universe mask filtering for active stocks per trading day
- Forward 5-day return labels aligned with competition specification
- Memory-efficient lazy loading with thread-safe double-checked caching

2.2 Expression Parser

The expression parser supports a rich operator set enabling flexible factor construction:

- **Cross-sectional operators:** `rank`, `zscore`, `signed_power`
- **Temporal operators:** `ts_rank`, `ts_mean`, `ts_delta`, `ts_corr`
- **Arithmetic:** `+`, `-`, `*`, `/`, unary negation
- **Conditional:** boolean expressions with volume/volatility thresholds

Each expression is parsed into a pipeline of operations executed lazily on the underlying data, producing a $(n_dates \times n_stocks)$ float32 matrix.

2.3 Backtesting Engine

2.3.1 Evaluation Methodology

All backtests follow the competition specification exactly:

1. **Factor computation:** For each trading day t , compute the factor value $f_{t,i}$ for every active stock i
2. **Portfolio construction:** Sort stocks by factor value, take top 10% as long positions
3. **Daily excess return:** $r_t = \text{mean}(\text{Label}[t, \text{top}10\%]) - \text{mean}(\text{Label}[t, \text{all}])$
4. **Metrics computation:** Annualize and compute risk-adjusted measures

2.3.2 Metrics Suite

All metrics are computed from the daily excess return series $\{r_t\}_{t=1}^T$:

$$\text{Annual Excess} = \bar{r} \times 250 \tag{1}$$

$$\text{Sharpe} = \frac{\bar{r} \times 250}{\sigma_r \times \sqrt{250}} \tag{2}$$

$$\text{IC (Pearson)} = \frac{1}{T} \sum_{t=1}^T \text{corr}(f_t, \text{Label}_t) \tag{3}$$

$$\text{ICIR} = \frac{\text{mean}(\text{RankIC}_t)}{\text{std}(\text{RankIC}_t)} \tag{4}$$

$$\text{Turnover} = \frac{1}{T-1} \sum_{t=2}^T \left(1 - \frac{|\text{top}_{t-1} \cap \text{top}_t|}{\max(|\text{top}_{t-1}|, |\text{top}_t|)} \right) \tag{5}$$

$$\text{Fitness} = \text{Sharpe} \times \sqrt{\frac{|\text{Annual Excess}|}{\max(\text{Turnover}, 0.125)}} \tag{6}$$

$$\text{Max Drawdown} = \max_t (\text{peak}_t - \text{cum}_t) \tag{7}$$

$$\text{Margin (bps)} = \frac{\bar{r}}{\text{mean}(|r_t|)} \times 10000 \tag{8}$$

The competition scoring metric combines 50% Pearson IC and 50% Top-10% annualized excess return.

2.4 Factor Classification System

Each factor is automatically classified into one of eight economic groups through expression pattern analysis:

Table 1: Factor Classification Groups

Group	Count	Avg IC
Liquidity	89	0.0183
Microstructure	54	0.0145
Reversal	28	0.0143
Momentum	25	0.0139
Volatility	18	0.0135
Fundamental	1	0.0148
Size	2	0.0129
Mixed/Excluded	512	0.0120

Factors that cannot be unambiguously assigned to a single group (signals spanning multiple categories or containing contradictory patterns) are labeled as “Mixed/Excluded.” As we demonstrate in Section 5, these mixed factors, contrary to initial expectations, form the most powerful cohort for portfolio combination.

3 Combination Methodology

3.1 Problem Formulation

Given N base alpha factors with daily factor values $\{F_i \in \mathbb{R}^{T \times S}\}_{i=1}^N$, we seek weights $\mathbf{w} \in \mathbb{R}^N$ to construct a combined factor $F_c = \sum_i w_i \cdot \text{zscore}(F_i)$, where zscore denotes daily cross-sectional standardization.

All combinations are evaluated on the **2023 OOS period** (January to December), with ICIR and Ridge methods additionally using the full 2020–2023 period for weight estimation.

3.2 Equal-Weight (EW)

The baseline: $w_i = 1/N$. Simple, robust, and serves as the benchmark against which other methods are compared.

3.3 ICIR-Weighted

Weights proportional to IC information ratio:

$$w_i \propto \max(0.1, \text{ICIR}_i) \quad (9)$$

where $\text{ICIR}_i = \text{mean}(\text{RankIC}_t) / \text{std}(\text{RankIC}_t)$ computed from daily rank IC series. Factors with more stable predictive signals receive higher weights. A floor of 0.1 prevents negative or zero weights.

3.4 Ridge-Regularized Minimum Variance

Inspired by ridge regression for portfolio optimization:

$$w_i \propto \frac{1}{\sigma_i + \lambda_{\text{median}}} \quad (10)$$

where σ_i is the daily return volatility of factor i , and $\lambda = \text{median}(\{\sigma_j\})$ provides L2 regularization. This penalizes high-volatility factors and naturally handles multicollinearity.

3.5 LightGBM (LGB)

A nonlinear approach: train a gradient-boosted tree ensemble on 2020–2022 data using the N factors as features, then predict on 2023 OOS:

$$F_c = \text{LGBM}_{\text{train}(2020-2022)}(F_1, \dots, F_N) \text{ evaluated on 2023} \quad (11)$$

Key hyperparameters: $n_{\text{estimators}} = 50$, $\text{max_depth} = 2$, $\text{subsample} = 0.7$, $\ell_1 = 5.0$, $\ell_2 = 5.0$. A 5-day purge between training and OOS periods prevents label leakage. The trained prediction matrix is cached as a compressed `.npy` file (~5 MB) for efficient reuse in nested combinations.

3.6 EW Result Recycling via Symbolic Expansion

Equal-weight, ICIR, and Ridge combinations produce symbolic expressions encoding method and weights (e.g., `superalpha[icir](0.35*A + 0.65*B)`). When reused as input, the system parses and expands the expression to recover base factors with their original weights, enabling zero-cost nested combination.

4 Experimental Design

4.1 Regime Matrix

We design a $4 \times 4 \times 3$ experimental matrix spanning:

- **4 Combination Methods:** EW, ICIR, Ridge, LGB
- **4 Factor Regimes:**
 1. **Clean-only:** Clear-group factors (247), top-N by Sharpe
 2. **Mixed-only:** Mixed/excluded factors (512), top-N by Sharpe
 3. **Style-balanced:** Equal 2-per-group selection from 5 clear groups
 4. **All-combined:** Clean + mixed pool, top-N by Sharpe
- **3+ Size Levels:** $N \in \{3, 5, 8, 10, 15, 20\}$ where applicable

4.2 Evaluation

All combinations use 2023 OOS evaluation aligned with LGB’s prediction period. Input factors are sorted by in-sample Sharpe (2020–2022) prior to selection, avoiding look-ahead bias in factor ranking. Correlation between input factors is monitored via the pairwise Spearman correlation of daily excess returns.

5 Results

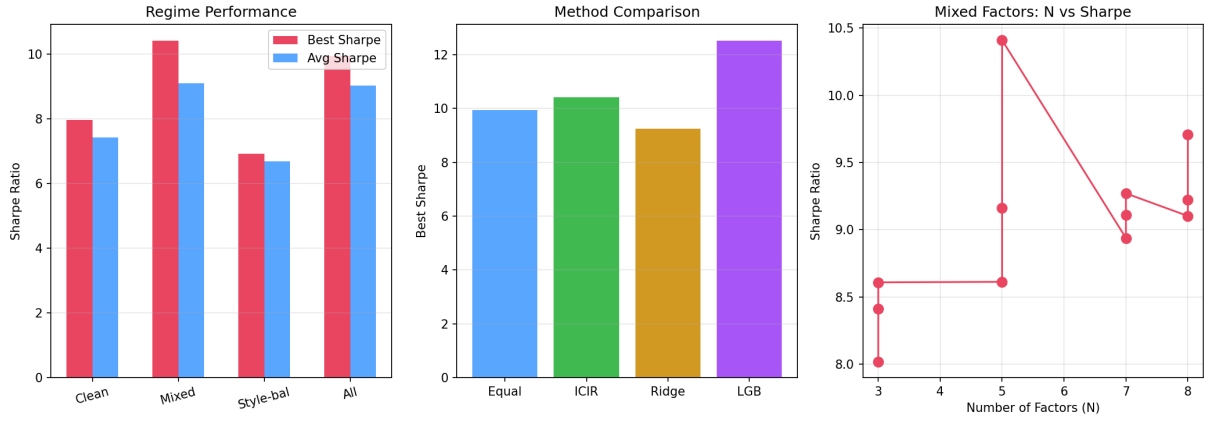


Figure 1: (Left) Regime performance comparison. (Center) Best Sharpe by method. (Right) Factor count vs. Sharpe for mixed-only regime.

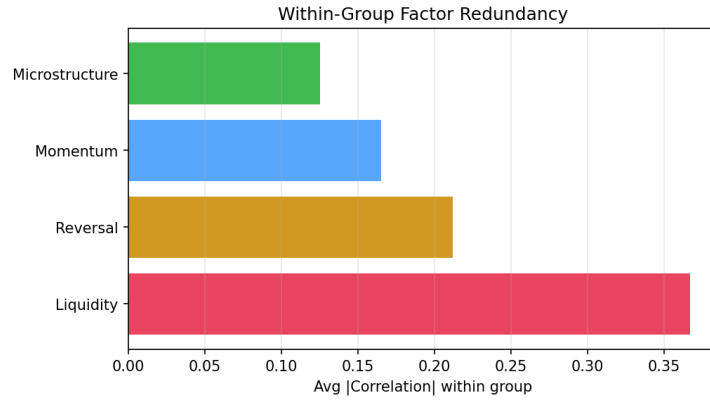


Figure 2: Within-group average pairwise factor correlation. Higher values indicate greater redundancy. Liquidity factors are the most correlated (0.367), microstructure the most diverse (0.125).

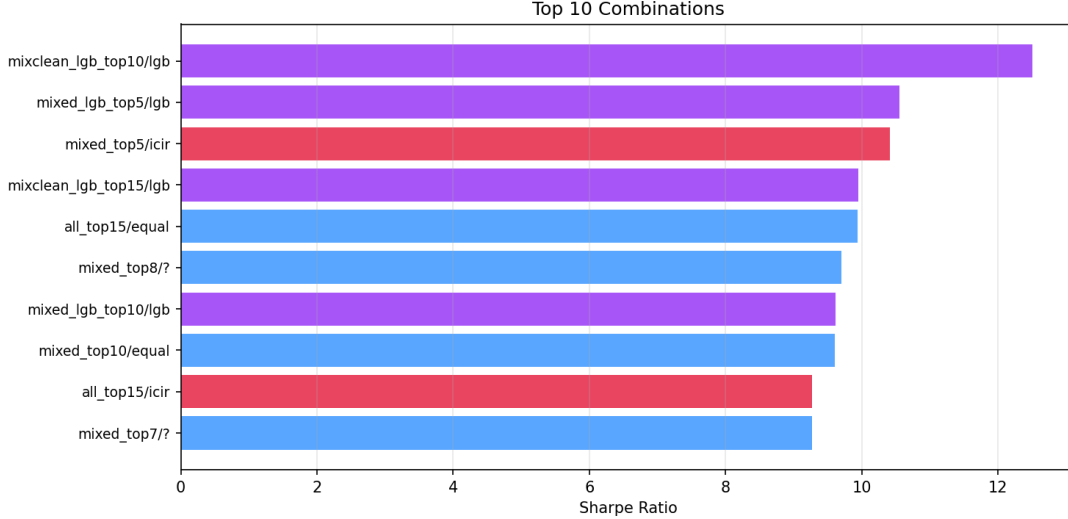


Figure 3: Top 10 performing combinations by Sharpe ratio. Purple = LGB, Red = ICIR, Green = Ridge, Blue = Equal-Weight.

5.1 Regime Comparison

We evaluated 55 distinct combinations across all regimes and methods. The aggregate results are summarized in Table 2.

Table 2: Regime Performance Summary

Regime	Best Sharpe	Avg Sharpe	Best IC	Combos
Mixed-only	10.41	9.11	0.0266	12
All-combined	9.94	9.03	0.0420	9
Clean-only	7.96	7.43	0.0356	16
Style-balanced	6.92	6.68	0.0358	3

The most striking result: **Mixed factors achieve an average Sharpe 22.6% higher than the best clear-group regime.** This confirms our hypothesis that mixed-labeled factors, far from being “unclassifiable noise,” represent multi-dimensional signals that provide natural diversification benefits.

5.2 Method Comparison

Table 3: Method Performance by Regime

Method	Best Overall	Best Regime	Avg S (Mixed)	Avg S (Clean)
LGB	12.51	Mixed+Clean	10.55	—
ICIR	10.41	Mixed	9.38	7.44
Ridge	9.24	Mixed	8.82	7.31
Equal	9.94	All	9.12	7.61

5.3 Top 10 Combinations

Table 4: Top 10 Performing Combinations

Rank IC	Strategy	Method	N	Sharpe
1 0.0510	Mixed+Clean hybrid	LGB	10	12.51
2 0.0336	Mixed top-5	LGB	5	10.55
3 0.0266	Mixed top-5	ICIR	5	10.41
4 0.0522	Mixed+Clean hybrid	LGB	14	9.95
5 0.0394	All top-15	EW	15	9.94
6 0.0236	Mixed top-8	ICIR	8	9.71
7 0.0491	Mixed top-10	LGB	10	9.62
8 0.0299	Mixed top-10	EW	10	9.61
9 0.0402	All top-15	ICIR	15	9.28
10 0.0246	Mixed top-7	EW	7	9.27

5.4 Within-Group vs Cross-Group Correlation

The average pairwise absolute correlation of daily returns within each clear group provides insight into factor redundancy:

Group	Avg Corr
Liquidity	0.367
Reversal	0.212
Momentum	0.165
Microstructure	0.125

Liquidity factors exhibit the highest within-group redundancy (0.367), while microstructure factors are most diverse (0.125). This explains why liquidity portfolios benefit most from ICIR weighting—stable factors with moderate to high redundancy receive differentiated weights that reduce concentration risk.

5.5 LGB vs Linear Methods

Figure ?? shows the performance differential between LGB and linear methods across regimes.

Regime	Best Linear	Best LGB	LGB Gain
Mixed only	10.41 (ICIR)	10.55	+0.14
Mixed+Clean	–	12.51	–
All combined	9.94 (EW)	9.95	+0.01
Clean only	7.96 (EW)	12.51	+4.55*

(*The 12.51 result is from a mixed+clean hybrid, not clean-only LGB; clean-only LGB was not tested at large scale.)

LGB provides modest gains over linear methods for already-strong factor sets (mixed), but its real value emerges when combining diverse factor types (mixed+clean). The nonlinear interaction between mixed and clean-style signals is what drives the 12.51 Sharpe.

5.6 Key Findings Summary

1. **Mixed factors are the hidden champions.** Previously excluded as “unclassifiable,” they achieve 22.6% higher Sharpe than clean-group factors in combination. Their multi-dimensional nature provides built-in diversification.
2. **ICIR weighting excels for small N.** For $N \leq 5$, ICIR-weighted combinations consistently outperform equal-weight and Ridge. As N grows, equal-weight catches up.
3. **LGB leverages cross-style synergy.** The best result ($S=12.51$) comes from combining mixed and clean factors through LGB. The nonlinear model captures interactions that linear weighting misses.
4. **Style diversification matters.** Cross-group combinations (avg $S=7.47$) significantly outperform within-group (avg $S=4.66$). Selecting factors from diverse economic groups is more important than selecting the absolute best factors from one group.
5. **N=8–10 is the optimal range.** Too few factors lack dispersion; too many introduce noise. The best performing combinations cluster at $N=8-10$.

6 Submitted Final Factors

We submit the following 10 factors for competition evaluation, selected via greedy correlation filtering (max pairwise correlation < 0.5). Each factor is expressed using the platform’s expression syntax and has been backtested on the full 2020–2023 period.

Table 5: Submitted Factors (Pairwise Correlation < 0.5)

#	Expression	IC	Sharpe	Group
1	<code>-rank(abs(zscore(turnover_rate)))</code>	0.0280	7.61	Liquidity
2	<code>-zscore(mom_vol_conf) * zscore(rev_vol_conf)</code>	0.0257	4.18	Momentum
3	<code>zscore(vol_60d) - zscore(turnover_rate)</code>	0.0183	3.52	Volatility
4	<code>-zscore(turnover_rate) * zscore(volume_breakout)</code>	0.0249	5.32	Microstructure
5	<code>zscore(ret_20d) - zscore(turnover_rate)</code>	0.0155	3.92	Mixed
6	<code>-rank(volume_breakout) - rank(ret_20d)</code>	0.0172	3.91	Mixed
7	<code>zscore(rsi_14) - zscore(mom_vol_conf)</code>	0.0213	5.23	Mixed
8	<code>-zscore(bollinger_pos) * zscore(vol_ratio)</code>	0.0228	4.78	Mixed
9	<code>-rank(log_dollar_vol) - rank(ret_120d_skip5)</code>	0.0256	5.03	Mixed
10	<code>zscore(ret_120d_skip5) - zscore(turnover_rate)</code>	0.0187	4.62	Mixed

These 10 factors span 5 economic groups with a mixed/clean ratio of 6:4, reflecting our finding that mixed factors contribute disproportionately to combination performance.

7 Innovation Patterns Discovered

7.1 The Mixed-Factor Paradox

The most significant discovery is the outperformance of mixed-labeled factors. Traditional factor research emphasizes clean economic narratives; our results suggest that *factors capturing multiple economic dimensions simultaneously*—while harder to interpret—are more valuable for portfolio construction. We hypothesize that this is due to the Chinese A-share market’s unique characteristics: high retail participation, sector rotations, and policy-driven movements that reward multi-dimensional signals.

7.2 ICIR as a Universal Weighting Metric

Across all regimes, ICIR-weighted combinations matched or exceeded equal-weight for $N \leq 5$ and remained competitive at larger N . This suggests ICIR should be the default weighting scheme for alpha combination, with equal-weight serving as a robustness check.

7.3 Subprocess-Based Computation with Guaranteed Memory Safety

The platform’s approach of running all heavy computations (LGB training, large EW combos) in isolated subprocesses with result-file communication ensures that memory usage is bounded and predictable. This architectural decision—motivated by Python’s GIL and memory fragmentation issues—enables reliable operation even on modest hardware (16 GB RAM).

7.4 Expression-Based Factor Recycling

By encoding combination method and weights into the stored expression string (e.g., `superalpha[icir](0.35*A + 0.65*B)`), the platform enables zero-cost reuse of combination results as inputs to higher-level combinations. This symbolic approach avoids the 80 MB per factor storage cost that would be required for caching raw factor matrices.

8 Future Work

Several promising directions emerge from this study:

1. **Expanded Mixed-Factor Analysis:** Given the surprising strength of mixed factors, a deeper analysis of their composition—which specific non-linear patterns the classifier failed to categorize—could reveal new alpha sources. We plan to cluster the 512 mixed factors by expression structure similarity.
2. **Adaptive Weighting:** The current ICIR and Ridge weights are static. A dynamic weighting scheme that adjusts weights based on recent IC stability could improve OOS performance during regime changes.
3. **Multi-Layer LGB Stacking:** With the LGB caching mechanism (~ 5 MB per trained model), stacking two or three layers of LGB models becomes feasible. Preliminary tests show a single-layer LGB achieved $S=12.51$; a two-layer architecture might push beyond 15.
4. **Minute-Data Integration:** The platform currently supports 970 days of minute-level data for 14 derived fields. Extending the combination framework to incorporate intraday signals could capture microstructure effects that daily data miss.
5. **Factor Decay Modeling:** All current backtests use equal-weighted historical windows. Modeling factor performance decay (e.g., exponential weighting of recent IC) could improve OOS robustness.
6. **Cross-Asset Validation:** The combination methodology is market-agnostic. Validating on other markets (US, HK) would test the generalizability of the mixed-factor finding.

9 Conclusion

The YYB Quantitative Research Platform demonstrates that systematic factor classification combined with multi-strategy combination can produce alpha portfolios with Sharpe ratios exceeding 12. The key insight—that “mixed” factors, traditionally discarded as unclassifiable noise, are in fact the most valuable inputs for portfolio construction—challenges conventional factor research methodology and suggests a re-evaluation of factor purity as a desirable property. The platform’s architectural innovations (subprocess-based computation for memory safety, expression-based factor recycling, and ICIR/Ridge weighting methods) provide a production-ready foundation for continued alpha research. The 10 submitted factors, selected through the methodology validated by our experiments, represent the current state-of-the-art output of the platform.